

The GTE–Wolfram Bridge

Rule 110, the Causal Graph, and the Selection Principle

novaspivack.com/research P48 complete theory monograph

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UGP Physics Tutorial Series

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UGP Physics Tutorial Series

Prerequisites (read first): *The GTE Polynomial: A Step-by-Step Cheat Sheet* · *How the UWCA Works* · *The Three-Tape CMCA*

Next tutorials: *Falsifiability: How to Test the GTE Framework*

See also: *The Levels of the Theory* · *How MDL Selects the 19-Bit Polynomial* · *Perfect Self-Containment and MDL*

Claim-Strength Taxonomy

Throughout this tutorial, results are labeled with a confidence grade:

- **A_{Lean} (CatAL)** — Machine-certified in Lean 4 with zero `sorry` and zero custom axioms. Independently verifiable by anyone with Lean installed.
- **A/D (CatAD)** — Analytically derived with full derivation, computationally confirmed. Not yet machine-certified.
- **A (CatA)** — Analytically derived; derivation complete but not independently machine-certified.
- **A/D (conditional)** — Derived under a named, stated premise; the premise is itself an open problem or a CatB assumption.
- **B (CatB)** — A stated working hypothesis or structural premise; not yet derived.

1 The Wolfram Physics Programme

What this tutorial addresses

The Wolfram Physics Project seeks a fundamental computational rule that generates spacetime and matter. GTE provides what Wolfram Physics lacks: a *selection principle* that identifies the unique such rule from first principles, and machine-certifies it. This tutorial explains the bridge between the two programmes.

1.1 The Wolfram Physics Approach

The **Wolfram Physics Project** (Stephen Wolfram and Jonathan Gorard, 2020) is the most recent systematic attempt to derive physics from a discrete computational rule. Its three key components are:

1. **The hypergraph rule.** The universe is modeled as a hypergraph (a collection of points with multi-way connections) that evolves by a local rewriting rule. Each application of the rule creates new connections and deletes old ones.
2. **The causal graph.** The **causal graph** records which rule applications causally depend on which others — essentially, which events produce the inputs for which later events. In the continuum limit, the causal graph encodes a Lorentzian metric on spacetime.
3. **Multway systems.** When multiple rule applications are possible, the **multiway system** tracks all of them simultaneously. Quantum mechanics is proposed to emerge from the structure of paths through the multiway graph.

The programme has produced a consistent framework for how spacetime geometry and general relativity can emerge from a discrete computational rule. Gorard and Wolfram have shown that Einstein’s equations appear in the continuum limit of appropriate causal graphs [1].

1.2 The Central Gap: Which Rule?

The Wolfram Physics programme is maximally inclusive: it defines the **Ruliad** as the union of all possible computations. The physical universe is somewhere in the Ruliad; the question is *where* — which specific rule?

The Wolfram approach provides no answer to this. The universe is not in the Ruliad because of some meta-selection process; it simply is where it is. This leaves the selection question open, which is the same open question that all of digital physics has faced since Konrad Zuse first proposed computational physics in 1969.

GTE addresses this gap directly.

2 What GTE Adds: The Selection Principle

GTE’s answer to “which rule?”

The unique MDL-minimal computational rule consistent with the PSC constraints is $p(L, C, R) = C + R - CR - LCR$ over $\text{GF}(7)$. This rule is selected not by the observer’s location in a Ruliad, but by an objective mathematical criterion: Minimum Description Length.

GTE (Generative Triple Evolution) applies two interacting selection criteria:

PSC (Perfect Self-Containment): The candidate rule must be the unique rule that can self-referentially justify its own selection from all possible rules. PSC is a set of five axioms about which theories can be considered complete without external input. It eliminates all but a vanishing fraction of possible rules.

MDL (Minimum Description Length): Among all PSC-admissible rules, the one with the shortest complete description is selected. MDL is the operational expression of **PI** (Presentation Invariance), the fifth PSC axiom.

Together, PSC and MDL uniquely identify the GTE polynomial $p(L, C, R)$ over $\text{GF}(7)$ as the rule governing our universe. This is not a heuristic argument — it is a derivation chain, with the key steps certified in Lean 4.

What this gives Wolfram Physics: A derivation, not a search. Instead of exploring the space of rules and comparing to observation, GTE proves which rule is unique and then verifies it matches observation.

3 The Triangle: Two Paths to Rule 110

The structure of GTE contains a remarkable internal triangle connecting three objects through two independent paths, both converging on Rule 110.

3.1 The Two Objects

- **The UGP arithmetic cascade.** The GTE framework starts from the UGP (**U**niverse **G**enerating **P**olynomial) arithmetic — a specific sequence of integer triples generated by the cascade map T , starting from the Lepton Seed $(1, 73, 823)$. This cascade is derived from the $N_{\text{gen}} = 3$ and $N_{\text{fam}} = 5$ generation/family counts.

- **The GTE polynomial p .** The polynomial $p(L, C, R) = C + R - CR - LCR$ over $\text{GF}(7)$ — the algebraic certificate of the GTE framework.

These look like independent objects. But they are not: there are two distinct paths from UGP to p , both ending at exactly Rule 110.

3.2 Path A: Direct-Interpolation Lift

Direct-Interpolation Lift Theorem (CatAL)

Statement: The total-parity shadow of the canonical cascade triples constrains all 8 binary input-output pairs of the GTE polynomial. Combined with MDL minimality (which enforces the vacuum-transparency condition), the unique multilinear rule consistent with these constraints is exactly $p(L, C, R) = C + R - CR - LCR$ over $\text{GF}(7)$. Rule 110 is a corollary (the binary restriction of p).

Machine-certified: LT-088-65–69 (CatAL, zero sorry, zero custom axioms).

The derivation in four steps:

Step 1 — The parity shadow. Reduce every coordinate of every cascade triple modulo 2 — keep only whether each value is even (0) or odd (1). This produces a sequence of binary triples. Collecting the unique ones from the ten canonical cascade triples gives 7 distinct binary (L, C, R) patterns, each paired with an output bit also read from the cascade.

Step 2 — The 8th row. One binary input pattern is not covered by the parity shadow: $(1, 1, 1)$ (all three neighbors odd). The vacuum-transparency condition — $p(0, 0, 0) = 0$ (the all-zeros state is a fixed point) — forces this remaining output to be 0. Now all 8 binary rows are specified.

Step 3 — Polynomial interpolation. A multilinear polynomial of degree ≤ 3 over three binary variables has exactly $2^3 = 8$ basis monomials: $\{1, L, C, R, LC, LR, CR, LCR\}$. With 8 known input-output pairs and 8 unknowns, the linear system $\mathbf{V} \mathbf{c} = \mathbf{y}$ over $\text{GF}(7)$ has a unique solution (the Vandermonde-type matrix is full rank).

Step 4 — The result. The unique solution is exactly $p(L, C, R) = C + R - CR - LCR$ over $\text{GF}(7)$, and the binary restriction $p|_{\{0,1\}^3}$ produces the Rule 110 truth table.

Worked numerical check: For input $(L, C, R) = (0, 1, 1)$: $p = 1 + 1 - 1 \cdot 1 - 0 \cdot 1 \cdot 1 = 1$. Rule 110 for input $011_2 = 3$: output bit is 1. ✓

3.3 Path B: MDL Direct Selection

Path B is the standard MDL derivation. Among all $k = 7$, $r = 1$ cellular automaton rules consistent with the PSC constraints (Standard Model derivability criterion), the shortest description is the GTE polynomial, requiring exactly 19 bits:

- 8 bits — the Rule 110 truth table (binary restriction)
- 3 bits — the prime modulus ($\lceil \log_2 7 \rceil = 3$)
- 5 bits — the monomial selector (which 4 of 8 basis terms are nonzero)
- 1 bit — chirality
- 2 bits — the -1 coefficient signs

A naive lookup table requires $343 \times \lceil \log_2 7 \rceil \approx 963$ bits — about 50 times larger. The MDL Uniqueness Theorem proves no shorter valid specification exists (CatAL) [2].

3.4 Convergence: Both Paths Identify Rule 110

The Direct-Interpolation Lift (Path A) shows that the UGP orbit’s arithmetic structure forces the Rule 110 truth table on the polynomial’s 8 binary rows. MDL selection (Path B) then forces the unique multilinear extension to all 343 GF(7) inputs. *Rule 110 is a theorem of GTE, not a postulate.*

This is the triangle: UGP arithmetic $\rightarrow$ orbit parity shadow $\rightarrow$ 8 binary rows $\rightarrow$ (by MDL interpolation) $\rightarrow$ unique polynomial $p \rightarrow$ (binary restriction) $\rightarrow$ Rule 110. The path is sequential and fully derived.

4 The Chirality Census: {Rule 110, Rule 124}

Chirality Census Theorem (CatAL)

Over all 120 orderings of the three SM fermion generations, the survivor set of rules consistent with the GTE orbit parity shadow is exactly {Rule 110, Rule 124} — the chiral pair related by left-right reflection. The choice of generation ordering convention is the *chirality gauge*: it selects one of the two chiral orientations but does not change any physical observable.

Rule 110 and Rule 124 are related by the AGL(1,7) reflection symmetry of the GF(7) alphabet (the map $x \mapsto -x \pmod{7}$). Machine-certified: `orbit_chirality_census (CatAL)` [1].

Why this matters for Wolfram Physics: The Wolfram Physics programme works with causal graphs that are directional. The left-right orientation of a CA rule is precisely the chirality that GTE identifies as a gauge degree of freedom. The fact that GTE derives exactly two chiral orientations (not one, not many) connects naturally to the chiral structure of the Standard Model’s weak interactions.

Worked example. Rule 110 has the bit pattern $01101110_2 = 110_{10}$ (reading binary outputs from (1,1,1) to (0,0,0) in order). Rule 124 has the bit pattern $01111100_2 = 124_{10}$. One is the left-right reflection of the other over the $\{0,1\}$ alphabet. Both are derived from the UGP orbit; neither is independently postulated.

5 The Causal Graph: 1023 Nodes and Queen Anne’s Lace

5.1 Generating the Causal Graph

The Wolfram Physics Project provides a framework for computing the causal graph of a hypergraph rule. For GTE, the relevant rule is `ruleGTE`: a WolframModel hyperedge rule encoding two overlapping CA triplet neighborhoods:

$$\{a, b, c\}, \{c, d, e\} \rightarrow \{a, b, f\}, \{f, c, d\}, \{e, b, g\}, \{g, d, h\}. \quad (1)$$

Running WolframModel for 10 generations produces a causal graph with **1023 nodes and 2044 edges** — a perfect binary tree of depth 10 [1]. Machine-certified:

- `perfectTree_numNodes`: $2^n - 1$ nodes by induction (CatAL).
- `gte_rulegte_ten_generations`: 1023 events at depth 10 (CatAL).
- `perfectTree_horton_ratio`: Horton ratio $r_B = 2$ at all levels (CatAL).

5.2 The Queen Anne’s Lace Structural Equivalence

The GTE causal graph (10 generations, 1023 nodes) is structurally equivalent to the Lindenmayer system $F \rightarrow F[+F][-F]$ at $\theta \approx 25^\circ$, $r \approx 0.70$ — the canonical L-system model of *Daucus carota* (Queen Anne’s lace, a flowering plant in the Apiaceae family) [1].

Five independent metrics agree:

- Horton branching ratio $r_B = 2$ (**CatAL**)
- Strahler number 10
- Fractal dimension $D \approx 1.943$
- Depth 10
- Bilateral symmetry

The shared computational primitive: **ruleGTE** creates 2 overlapping pairs from 1 (binary branching), exactly as meristematic apical cell division creates 2 daughter cells. This is a structural equivalence (**CatA**) — a shared graph topology — not a formal isomorphism between cellular automaton dynamics and plant growth. The physical significance is that the GTE causal graph’s fractal structure has the Horton ratio $r_B = 2$ machine-certified as a theorem of the rule’s combinatorics, not as a measured fit.

6 Three Key Distinctions for Wolfram Physics Readers

Readers familiar with the Wolfram Physics Project should note three precise differences between GTE and the Wolfram approach.

6.1 Distinction 1: The CA Is the Certificate, Not the Substrate

The most important difference. The Wolfram Physics Project proposes that the universe *is* a hypergraph rule evolving on a hypergraph.

GTE proves this cannot be exactly right: no finite cellular automaton can be the exact physical substrate of our universe.

Machine-certified: `no_finite_ca_exact_lorentz_replica` (**CatAL**, zero sorry) [3].

The reason: every CA with a finite lattice spacing has Lorentz-violating artifacts at that spacing. The physical substrate must recover exact Lorentz invariance, which no lattice CA achieves.

The GTE resolution is the two-level architecture:

- **Level 1 (certificate):** The CMCA (the discrete CA) encodes the structural constraints of the physical field via the Algebraic Lifting Theorem (**CatAL**). It is an algebraic certificate, not the substrate.
- **Level 2 (substrate):** The Φ_{MDL} field is the actual physical substrate — a continuous $\mathbb{Z}_7$ -symmetric Klein-Gordon field on $\mathbb{R}^{3+1}$.

GTE does not claim the universe IS a CA. It proves the universe’s physics has an algebraic certificate that IS a CA. These are different claims, and the distinction matters enormously for Lorentz invariance and quantum mechanics.

6.2 Distinction 2: MDL Uniquely Selects the GTE Polynomial

The Wolfram Physics Project explores the space of rules computationally and compares to observation. GTE selects the rule analytically.

The polynomial $p = C + R - CR - LCR$ over $\text{GF}(7)$ is the unique MDL-minimal description consistent with the PSC constraints — machine-certified in Lean 4, zero sorry [2]. Every $\mathbb{Z}_N \times \mathbb{Z}_M$ competitor has been enumerated and eliminated. MDL is the selection principle that digital physics has lacked.

This means GTE does not search for rules; it derives the rule.

6.3 Distinction 3: Rule 110 Is a Theorem, Not a Postulate

In Wolfram’s programme, Rule 110 was discovered by computational enumeration and its Turing universality proved by Matthew Cook (1994, published 2004). It is an empirical discovery about which simple CA is computationally interesting.

In GTE, Rule 110 is a theorem:

$$p|_{\{0,1\}^3} = \text{Rule 110},$$

machine-certified: `rule110_z7_poly_rep (CatAL, zero sorry, native_decide)` [2]. Rule 110 is the binary restriction of the GTE polynomial, derived from the UGP orbit arithmetic. Cook’s proof of Rule 110’s Turing universality follows as a structural consequence (`phimdl_turing_universal, CatAL`) [3].

7 The 19-Bit Program: Connecting MDL to Wolfram Language

A central claim of GTE is that the universe’s complete algorithmic description requires **19 bits**. This is not an approximation; it is the MDL complexity of the GTE polynomial in its natural description class.

7.1 The 19-Bit Decomposition

Table 1: The 19-bit decomposition of the GTE polynomial description. Each bit count is exact and follows from the structure of the polynomial representation class.

Component	Bits	What it encodes
Rule 110 truth table	8	Binary outputs for all 8 binary inputs
Prime modulus	3	$\lceil \log_2 7 \rceil$ — selects the $\text{GF}(7)$ alphabet
Monomial selector	5	Which 4 of 8 basis monomials are nonzero
Chirality	1	Left-right orientation (the chirality gauge)
Coefficient signs	2	The -1 coefficients on CR and LCR
Total	19	The complete GTE polynomial

For comparison, specifying the rule as a lookup table requires $343 \times \lceil \log_2 7 \rceil = 343 \times 3 \approx 963$ bits. The polynomial representation is $963/19 \approx 50\times$ more compressed.

7.2 Wolfram Language Implementation

The three-tape CMCA that implements the GTE polynomial can be written directly in Wolfram Language using `CellularAutomaton` and `WolframModel`. The rule `ruleGTE` produces the causal graph discussed in §5. The polynomial’s 343 outputs can be verified by:

```
Table[Mod[C + R - C*R - L*C*R, 7], {L,0,6}, {C,0,6}, {R,0,6}]
```

This generates the complete GTE update table from the polynomial formula. The Rule 110 connection is verified by restricting to $\{L, C, R\} \in \{0,1\}^3$: the resulting 8 outputs match the Rule 110 truth table exactly.

8 Open Questions at the Bridge

Several questions at the GTE–Wolfram interface remain open. Presenting them honestly is essential to the scientific record.

8.1 The Causal-Graph / Gorard Chain Correspondence

The Wolfram Physics programme identifies the emergent spacetime metric with the limiting structure of the causal graph. Gorard’s result shows that Einstein’s equations appear in the continuum limit of appropriate causal graphs.

GTE uses a related object: the **Gorard chain** (defined in P48) is a sequence of causal events in the Φ_{MDL} field. The *precise correspondence* between the P48 Gorard-chain limit and the Wolfram Physics causal graph is stated as an open question in the GTE framework. Both directions — GTE’s and Wolfram’s — point toward the same Einstein equations; whether they can be formally identified as the same limiting structure is not yet proven [1].

8.2 Quantum Mechanics from the Multiway System

The Wolfram Physics programme proposes that quantum mechanics emerges from paths through the multiway evolution graph. GTE derives quantum mechanics — including the Born rule (**CatAL**, **born_rule_unconditional**) — from the $[D]$ -weighted path integral over kink configurations.

Whether the GTE $[D]$ -weighting is the same mathematical object as the Wolfram-Gorard branch weights in the multiway graph is a precise open question at the interface of the two frameworks.

9 Summary: What GTE Offers Wolfram Physics

The five contributions

1. **A derivation, not a search.** GTE identifies the unique rule (the GTE polynomial over $\text{GF}(7)$) from first principles, via PSC and MDL. No exploration of rule space is required.
2. **Machine certification.** The key steps — MDL uniqueness, Rule 110 as a theorem, the chirality census — are Lean 4 certified, zero sorry. The derivation chain is formally verified.
3. **A 19-bit description.** The GTE polynomial is the minimum description of the universe’s computational rule. This connects Wolfram’s emphasis on simple rules to a precise bit-count theorem.
4. **The CA as certificate, not substrate.** GTE resolves the Lorentz-invariance tension in discrete physics by separating the algebraic certificate (the CMCA) from the physical substrate (the Φ_{MDL} field).
5. **A causal graph with certified structure.** The GTE causal graph from **ruleGTE** has machine-certified tree structure, Horton ratio, and an explicit structural equivalence to a botanical branching system.

Further Reading

Primary Sources

The results in this tutorial are derived in the following papers, all available open-access on Zenodo and at novaspivack.com/research:

- **P48 — The Complete GTE Framework** (main synthesis monograph, including the Wolfram Physics context chapter):
doi.org/10.5281/zenodo.20560550
- **P49 — MDL Selects the Wolfram Rule: $\mathbb{Z}_7$ Dynamics** (the GTE–Wolfram bridge and the Direct-Interpolation Lift Theorem):
doi.org/10.5281/zenodo.20572656
- **P28 — Computational Universality and the Standard Model** (no-CA-replica theorem; Turing universality from GTE):
doi.org/10.5281/zenodo.20259513
- **P45 — The Three-Tape Chiral Minkowski Cellular Automaton** (the CMCA as the Level-1 algebraic certificate):
doi.org/10.5281/zenodo.20465805

Lean 4 machine proofs: github.com/novaspivack/ugp-lean

References

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- [3] Nova Spivack. Computational universality and the standard model: Rule 110, $\mathbb{Z}_5$ rings, and mod-7 structure in the universal generative principle. Zenodo, 2026. URL <https://doi.org/10.5281/zenodo.20259513>. Paper P28 in the UGP Physics series. <https://doi.org/10.5281/zenodo.20259513>.